Final Project Essentials in Text and Speech Processing

Benchmarking LLMs and VLMs for Automated De-Identification of Clinical Text and Medical Images

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Project description

Protected health information appears both in narrative clinical text (names, dates, addresses) and directly into the pixels of medical images (patient name on X-rays or ultrasound frames). This project splits into two complementary tracks using the same base model family so that differences in de-identification performance can be attributed to modality rather than to differences between unrelated models. We propose the **Qwen** family, as it is licensed and freely available on Hugging Face without gated access. One team member benchmarks the text-only model on de-identifying clinical notes, while the other benchmarks the corresponding vision-language model on de-identifying PHI embedded in medical images (X-rays and similar).

Project Motivation & Added Value

Using one model family across both modalities lets us focus on a more specific question: does the same underlying model retain its de-identification ability when PHI moves from text tokens to pixels?. This matters in a practical deployment, because hospitals often need to redact PHI from both note text and image overlays/headers in the same pipeline, and a model strong on one but weak on the other might signify a "blind spot". Since a missed PHI mention is a real privacy leak we prioritize recall over precision in our evaluation design.

Expected Challenges

- Sourcing fully open-source datasets with ground-truth PHI annotations for both tracks, since most gold-standard clinical corpora, such as MIMIC, require a data use agreement. We will use the MIDI-B (MIDRC Medical Image De-Identification Benchmark) dataset for the image track, paired with an open synthetic or lightly-restricted corpus for the text track.
- Designing prompts that help obtain structured, scoreable PHI-span outputs consistently from both the LLM and the VLM.
- Implementing a recall-oriented evaluation design, so that rare PHI types are not masked by an aggregate F1 score.
- Keeping the two tracks comparable (same PHI categories, same scoring script, same base model family) despite being run by different team members.
- **Modality vs. dataset confound.** ASQ-PHI and MIDI-B differ in both modality and in the data/PHI types they contain, so a performance gap between tracks could reflect the dataset rather than the modality. We control for this by mapping both datasets' PHI categories onto one shared taxonomy (Step 1) and restricting the cross-modality comparison (Step 5) to the shared categories that have a real counterpart with enough support in *both* datasets, reporting per-category results so dataset-specific categories are never silently pooled into an overall score.
- **No external reference point.** Comparing only zero-shot and few-shot Qwen tells us how prompting affects the model, not how good the model is in absolute terms. We add a non-LLM

baseline for the text track (Step 2) so the LLM conditions are measured against an existing de-identification tool, not only against each other.

Planned Steps

1. **Dataset Sourcing & Preprocessing**
   - Confirm now, before Step 2/3 begin, that ASQ-PHI's and MIDI-B's licenses permit our use, including training/evaluating a local model, and redistribution of derived (non-PHI) scores and anonymized examples in the final report; document the exact license/terms for both in the taxonomy-mapping document below.
   - Download ASQ-PHI and MIDI-B.
   - Parse ASQ-PHI's JSON PHI tags into a shared schema.
   - Parse MIDI-B's ground truth into an analogous schema for images, separating DICOM-header PHI from pixel-burned-in PHI.
   - Define and document the shared PHI taxonomy (5–8 top-level categories) that both ASQ-PHI's 13 categories and MIDI-B's categories are mapped onto, and flag which shared categories exist in *both* datasets with enough support to be used in the cross-modality comparison (Step 5).
   - Outcome: two preprocessed, versioned benchmark files (JSON/CSV) in a common schema, plus a one-page taxonomy-mapping document that also records license terms and the both-datasets category overlap.
2. **LLM-Based Text De-Identification**
   - Serve Qwen3-8B-Instruct locally.
   - Write a structured extraction prompt: instruct the model to return a JSON array of {type, text} spans using the shared taxonomy, with 2–3 few-shot examples covering easy and tricky cases (e.g. relative dates, nicknames, partial addresses).
   - Use constrained JSON decoding (or Pydantic model) so outputs are always parseable, avoiding manual regex clean-up.
   - Run two prompting conditions: zero-shot and few-shot; optionally a third condition using Qwen3's "Thinking" mode as a small ablation on whether extra reasoning improves recall.
   - **Non-LLM baseline.** Run Microsoft Presidio (open-source, rule-based + spaCy-NER PII/PHI detector) on the same ASQ-PHI queries, with its output categories mapped onto the shared taxonomy, scored by the exact same pipeline as the LLM conditions. This gives zero-shot/few-shot Qwen an external reference point instead of only being compared to each other.
   - Outcome: one prediction file per condition (zero-shot, few-shot, optionally thinking, and the Presidio baseline) for the full ASQ-PHI set, ready for scoring.
3. **VLM-Based Image De-Identification**

- Serve Qwen3-VL-8B-Instruct the same way, feeding each MIDI-B image plus a prompt asking for PHI location and category.
- **Approach: OCR + classify.** First extract all burned-in text with an OCR step (Qwen3-VL's own text-reading ability, or a dedicated OCR pass), then classify each extracted text span into the shared PHI taxonomy using the same category logic as the text track. This keeps the image task close to a text-classification problem on OCR'd strings.
- Localization: each OCR'd string already carries its bounding box, scoring reduces to the same span-matching logic as the text track, applied to OCR output instead of raw text.
- For every MIDI-B image, a list of OCR'd text spans with bounding boxes and predicted PHI category, ready for scoring.

4. **Shared Evaluation Framework**
   - Implement a common evaluation pipeline for both modalities, scoring every text-track condition (zero-shot, few-shot, Presidio baseline) and the image-track VLM output the same way.
   - Measure precision, recall, and F1 score, with particular emphasis on PHI recall.
   - Report performance separately for different PHI categories to identify rare or difficult-to-detect information types.
   - Analyse false negatives and false positives to characterize common model failure modes.
5. **Cross-Modality Comparison**
   - Compare the LLM and VLM only on the shared-taxonomy categories confirmed in Step 1 to exist in *both* ASQ-PHI and MIDI-B with enough support, so a gap is not attributable to a category that only one dataset contains.
   - Within that controlled comparison, investigate whether certain types of PHI are systematically harder to detect in text or images.
   - Separately, report each track's performance against its own non-LLM baseline (Presidio for text; an existing image de-identification/OCR-redaction tool for images, if available) to contextualize modality differences against tool-level differences.
   - Analyse modality-specific failure cases and identify potential blind spots in the models.
   - Assess the implications of the results for automated clinical de-identification pipelines.
6. **Final Analysis & Reporting**
   - Summarize the key findings, limitations related to datasets, prompting, model capabilities, and generalizability.
   - Write the final report and formulate recommendations for future work.